ROBOTICS CLUB · RECRUITMENT 2026

RoboReach

Fresher Selection Tasks

Wireless, manually controlled robotic-arm bot · 7-minute, 15-obstacle arena

The RoboReachproblem statement asks teams to build a **manually controlled, wireless bot with a mounted robotic arm** that picks, carries, stacks, throws, draws and places objects across a 3×3 m arena. It has to be stable on ramps, precise at the gripper, safe, and fully fail-safe when the radio link drops. This selection round is your way into the team.

You are **not** expected to know everything. We want to see how quickly you learn, how you make decisions, and whether you can explain them.

Read this first

- **Time:** Tasks run from easiest to hardest, so attempt them in order. Partial answers earn partial marks.
- **Submit one GitHub repository** (folder structure in Task 5). The submission link and deadline will be announced by the club.
- **Justify every decision.** A number or a part choice with no reason earns very few marks.
- **Interview:** questions will be drawn directly from your submission. Be ready to explain and defend every figure, diagram, CAD feature and calculation.
- You may use tutorials, forums and AI tools to learn. **Cite every source** in your README. Work you cannot explain in the interview will be marked down.
- **Task 4 is a design challenge. Choose only ONE** of Challenge A or B based on what you enjoy most.

Point distribution

Task	Title	Who	Points
1	Quick Rulebook Check	Everyone	5
2	Mechanism Decision (drive, arm, gripper)	Everyone	15
3	Circuit Design & Power Budget	Everyone	25
4	Design Challenge (CAD): Challenge A: Stable Object-Carrying Bot **or** Challenge B: Gripper for a Misaligned Object	Pick one	45
5	Repository & Documentation	Everyone	10
	Total		**100**

Which challenge should I pick for Task 4?

Challenge	Choose it if you like…	Tools you can use
A · Stable Object-Carrying Bot	Chassis and drive design, centre-of-gravity and incline calculations, assembling a complete robot in CAD.	CAD: Fusion 360, SOLIDWORKS, Onshape or FreeCAD. Sheets: Excel/Google Sheets. Animation: Fusion Animation, Motion Study or Blender
B · Gripper for a Misaligned Object	Linkages and mechanisms, compliance, force and torque calculations, clever small-scale design.	Same tools as Challenge A

TASK 1 Quick Rulebook Check — 5 pts

Skim the RoboReachrules (summary in Appendix B) and answer briefly. One or two lines per item is enough.

- **(1 pt) Size and reach.** What is the start-box size, at what height must the arm grasp an object, and what is the minimum ground clearance?
- **(1 pt) Disconnect.** What must happen to the motors, the arm and the gripper when the link to the joystick is lost?
- **(1 pt) Power.** The limit is 24 V DC between any two points. Show by calculation whether a fully charged 3S LiPo (4.2 V per cell) and a fully charged 6S LiPo comply.
- **(2 pts) Score calculation.** A bot completes Pickup, Ascent, Bridge, Descent, Touchdown and The End. It finishes in 250 s, drops the cube once after pickup, and one wheel leaves the track once. Compute A, T, P and the total score, showing each step.

Deliverable

`docs/task1_rulebook.md` (or PDF): numbered answers matching the items above.

TASK 2 Mechanism Decision — 15 pts

A good robot starts with good decisions. For each part below, **compare at least three options with a weighted decision matrix** (criteria, weights summing to 100%, scores from 1 to 5, weighted totals), pick a winner, and write one sentence on the runner-up and why it lost. Use the arena facts in Appendix B. Hand sketches are fine.

2a. Drive configuration (5 pts)

- Options to consider: 2-wheel differential with a caster, 4-wheel skid-steer, mecanum/omni wheels, tracks.
- Criteria drawn from the arena: climbing the 15° ramp, crossing 40 mm speed breakers, turning inside a 250 mm-wide track, stability with a front-heavy arm, traction, cost and complexity.
- Top-view sketch of your chosen layout with wheelbase, track width and wheel diameter. Check that your wheel diameter can clear a 40 mm step.

2b. Arm mechanism (6 pts)

- The arm must reach an object whose centre is 450 mm above the ground, keep ground clearance ≥ 50 mm, and fold into a $300 \times 200 \times 300$ mm box at the start.
- Options to consider: 2-link revolute arm, 3-link revolute arm with wrist, linear lift (lead screw or scissor) with a short arm, telescopic arm.
- Criteria: reach-to-stowed ratio, number of actuators (degrees of freedom), torque on the base joint, stiffness and backlash, precision for tasks like Buzzed and Robonardo DaVinci, complexity.
- Side-view sketches of your chosen arm **stowed** and **extended**, with approximate link lengths. Verify the 450 mm reach with simple geometry, for example $z = h_s + L_1 \sin\theta_1 + L_2 \sin(\theta_1 + \theta_2)$.

2c. Gripper mechanism (4 pts)

- The gripper must handle 50 mm and 60 mm cubes, a $\varnothing 60 \times 80$ mm cylinder, a cone, a marker and a loop with a handle. It must be purely mechanical (no adhesives, magnets, hooks or suction).
- Options to consider: parallel-jaw gripper (rack and pinion or lead screw), two-finger linkage gripper, underactuated adaptive fingers, scoop or fork.
- Choose one, sketch it, and say which actuator drives it and how the object is kept from slipping.

Deliverable

`docs/task2_mechanism/`: the three decision matrices, sketches and short justifications.

TASK 3 Circuit Design & Power Budget 25 pts

Design the complete electrical system of a RoboReachbot. Appendix A gives reference ratings and a pin map; for your own parts, use the real datasheet or supplier values and say where they differ.

3a. Schematic and wiring diagram (9 pts)

- Labelled diagram showing: battery, kill-switch, fuse, buck converters, controller, motor driver, 2 drive motors, 5 servos and the wireless receiver or module.
- Mark every supply voltage and the **common-ground** topology. Show which rail powers what.
- Explain why the servos should not be powered from the controller's 5 V pin, what the motor driver does functionally, and where you would place decoupling capacitors and why.
- State the wire gauge you would use for the high-current paths (battery to driver, battery to buck converter) and why.

Tools: KiCad, Tinkercad Circuits, draw.io, or a clear photographed hand drawing.

3b. Electronics BOM, power budget & battery (8 pts)

- Electronics-only BOM within **INR 12,000 including shipping and about 10% contingency**: exact model, quantity, unit price, supplier link, key specs and one line of rationale for each item.
- Table of idle and peak (stall) current for every powered component, grouped by voltage rail, with the worst-case total. Size each buck converter with at least $1.5\times$ margin over its worst-case load.
- Estimated runtime for your battery (mAh, V) over a 7-minute run plus testing. State your assumptions: what counts as a "typical" load, and what usable fraction of capacity you assume.
- Compare two chemistries (e.g. LiPo vs. Li-ion 18650) on energy density, C-rating, weight and cost, and justify your pick. Confirm the ≤ 24 V rule, the kill-switch current rating, and that the battery cannot detach if the bot tips over.

3c. Datasheet check (4 pts)

- Pick one active component (motor driver, servo or buck converter). Name two datasheet or supplier specs that directly constrained your design and say how each shaped your circuit or BOM.
- Name one condition or rating that would **damage** the component if violated, and explain how your design avoids it.
- *Hint:* compare the drive-motor stall current in Appendix A with the rating of the driver you selected. What do you do about any mismatch?

3d. Troubleshooting (4 pts)

Pick **one** fault: (a) the controller resets when the shoulder servo starts moving; (b) the servos jitter randomly; (c) one drive motor only turns one way; (d) the phone connects but the bot does not respond. For your fault give the probable root causes, the **first** diagnostic check you would perform, and how you would confirm the fix.

> **Deliverable**
> `docs/task3_circuit/`: schematic, electronics BOM, power budget and chemistry comparison, datasheet check, troubleshooting analysis. No code required.

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TASK 4 Beginner CAD Design Challenge (choose ONE) 45 pts

Choose **only ONE** of the following challenges. You do not need prior experience with advanced CAD tools or simulation software. Focus on a simple, workable design, basic engineering calculations and explaining your decisions.

You may use Fusion 360, SOLIDWORKS, Onshape or FreeCAD. Use standard components where possible. Approximate dimensions are acceptable if you clearly state your assumptions.

Challenge A — Design a Stable Robot Chassis

Problem statement

Design a simple mobile robot that can carry a cube using a front-mounted gripper. The robot should be stable, compact and capable of moving on a flat surface and a small ramp without tipping over.

Design specifications

- **Chassis:** Design a chassis with four wheels. Choose reasonable dimensions for length, width and height.
- **Gripper:** Add a simple front-mounted gripper or a basic two-finger holding mechanism. The gripper may be modelled using simple blocks and joints.
- **Payload:** Include a 50 mm cube in the CAD assembly. Its centre should be approximately 70 mm ahead of the front edge of the chassis.
- **Compactness:** The robot must fit inside a $300 \times 200 \times 300$ mm box at the start.
- **Stability:** Explain how your chassis dimensions and payload position help prevent the robot from tipping. Consider a ramp of up to $15°$.

What you need to do

- Create a 3D CAD model of the chassis, four wheels, gripper and cube. You may use simple shapes instead of detailed motor and gearbox models.
- Choose a wheel diameter, wheelbase and track width. Explain your choices in a few sentences.
- Estimate the combined centre of gravity of the robot and cube using reasonable assumed masses and dimensions. Show the basic calculation.
- Explain why your robot should remain stable on a ramp. A detailed simulation or stress analysis is not required.
- Create one exploded view and a short animation showing the robot moving forward and carrying the cube. A simple assembly motion is sufficient.

Deliverables

- 3D CAD assembly with chassis, wheels, gripper and cube.
- One labelled drawing showing the main dimensions.
- A short stability calculation and a list of selected components.
- One exploded view and a short motion animation.

How it is marked (45 pts)

Criterion	Points
CAD model quality, completeness and dimensional clarity	16
Basic stability reasoning and payload placement	11
Component choices and engineering justification	9
Exploded view, animation and presentation	9

Challenge B — Design a Simple Self-Aligning Gripper

Problem statement

Design a simple mechanical gripper that can pick up a cube even when the cube is not perfectly centred in front of the gripper. The gripper should guide the cube into position and hold it securely while being lifted.

Design specifications

- **Target object:** Use a 50 mm cube as your main object.
- **Misalignment:** Your gripper should accommodate a sideways offset of up to ± 10 mm and an angular misalignment of up to $\pm 10°$.
- **Gripping mechanism:** Use two moving fingers with angled or slightly funnel-shaped inner surfaces to guide the cube

towards the centre.
- **Actuation:** Model one servo motor or another simple actuator. You do not need to build a working electronic circuit.
- **Contact surfaces:** Explain how the finger surfaces prevent slipping without damaging the object.

What you need to do

- Create a 3D CAD model containing a mounting plate, two fingers, a simple linkage or sliding mechanism, an actuator model and the target cube.
- Show how the fingers open and close. You may use simplified joints and motion constraints.
- Demonstrate the gripper in two cases: a centred cube and a cube placed slightly off-centre.
- Estimate the force needed to hold the cube. Use the cube's assumed mass and explain the role of friction. A detailed finite-element analysis is not required.
- Choose an actuator and explain why it is suitable. You may use a real product datasheet or clearly state reasonable assumed specifications.

Deliverables

- Complete CAD assembly of the gripper and target cube.
- One labelled drawing showing the main dimensions and mounting points.
- A basic grip-force calculation and a short list of selected components.
- One exploded view and a short animation showing centred and misaligned grasping.

How it is marked (45 pts)

Criterion	Points
Gripper CAD quality, completeness and mounting design	11
Ability to accommodate the specified misalignment	14
Basic force calculation and component justification	11
Exploded view, animation and presentation	9

Beginner-friendly guidelines

- You are not expected to model every screw, gear tooth, motor winding or electronic component. Simplified shapes are acceptable.
- You do not need to perform advanced simulation, inverse kinematics or finite-element analysis.
- Use realistic dimensions, label your assumptions and show your working for basic calculations.
- A simple design that works logically and that you can explain is better than a complicated design you cannot justify.
- You may use tutorials and AI tools for learning, but cite your sources and be prepared to explain your own CAD model during the interview.

TASK 5 Repository & Documentation 10 pts

- **(3 pts) Structure:** follow the folder layout below. Nothing duplicated across tasks.
- **(3 pts) README:** how to open every file, your assumptions, limitations, tools used and references.
- **(2 pts) Clarity:** written notes of about 2–4 pages in total (excluding BOM, drawings, tables and CAD). Short and precise beats long and vague.
- **(2 pts) Commit history:** at least 5 meaningful commits spread over the days. It shows how you worked.

Final deliverable: one GitHub repository

```
repo/
|-- README.md    assumptions, limitations, tools used, references, chosen challenge
|-- docs/        task1_rulebook.md
|                task2_mechanism/   decision matrices, sketches
|                task3_circuit/     schematic, BOM, power budget, datasheet check,
|                                   troubleshooting
|-- design/      Challenge A or B: CAD files (STEP + native), calculations,
|                BOM, drawings
‘-- media/       exploded-view and motion videos (or a link file if too large)
```

Videos larger than GitHub’s file limit can be uploaded to Google Drive; put a public link in `media/links.md`.

Optional extensions (bonus, used as tie-breakers)

- 3D-print or cardboard-prototype your gripper or chassis and share a one-minute video of it working.
- Run a simple stress or deflection simulation on a gripper finger or chassis member (Fusion, SOLIDWORKS or FreeCAD).
- Design the power-distribution board as a PCB layout in KiCad.
- A small firmware or Python sketch: joystick-to-motor mixing with dead-zone, or a 300 ms link watchdog that stops the motors and holds the gripper position on disconnect.
- Full RoboReacharm in CAD with a two-link inverse-kinematics check that reaches the 450 mm target.

> **Interview notice**
>
> - Every point you make in your submission is fair game. Expect questions such as “change this one value, what happens?”, “why not the other option?” and “walk me through this part of your CAD”.
> - We value **what you learned in two days** far more than a polished copy of a tutorial.
> - Marks are awarded for correctness (what), justification (why) and clarity (how well you explain it).

Appendix A — Reference Hardware Sheet (for circuit design)

Representative values for a typical build. For Task 3 you must use the real datasheet values of the parts **you** choose, and say where they differ from this sheet.

A.1 Controller & pin map: ESP32-WROOM-32 DevKit

Signal	GPIO	Function
ENA (PWM)	14	Left drive motor speed, 0–255
IN1 / IN2	27 / 26	Left motor direction
ENB (PWM)	25	Right drive motor speed, 0–255
IN3 / IN4	33 / 32	Right motor direction
Servo: base / shoulder / elbow / wrist	13 / 16 / 17 / 18	Arm joints, 50 Hz servo PWM
Servo: gripper	19	Gripper open/close
Status LED	2	Link state indication

GPIO 6–11 are wired to the module's flash memory and must **not** be used. Logic level is 3.3 V.

A.2 Power rails

Rail	Source	Feeds
11.1 V nominal (12.6 V full, ≈9.9 V recommended cut-off)	3S LiPo through the kill-switch	Motor driver supply; input of both buck converters
6.0 V	Buck converter, ≥ 5 A	All 5 servos
5.0 V	Second buck converter, ≥ 2 A	ESP32 Vin
3.3 V	ESP32 on-board regulator	Logic signals
GND	Common star point	Every ground must be tied together

A.3 Component reference ratings

Component	Qty	Reference ratings
12 V gearmotor, 100 RPM	2	No-load 0.25 A · typical running 1.0 A · **stall 4.0 A** · stall torque 18 kg·cm
Dual H-bridge driver (L298N-class)	1	2 A continuous per channel · ≈2 V drop · motor supply up to 35 V · logic 5 V (accepts 3.3 V control signals)
MG996R-class metal-gear servo	4	4.8–7.2 V · 9.4 kg·cm at 4.8 V, 11 kg·cm at 6 V · idle ≈10 mA · moving ≈500 mA · stall ≈2.5 A at 6 V · 55 g
MG90S-class micro servo (gripper)	1	4.8–6 V · 1.8 kg·cm at 4.8 V, 2.2 kg·cm at 6 V · stall ≈0.7 A · 13 g
ESP32 DevKit	1	Typical ≈80 mA, radio peaks ≈350 mA at 5 V input
3S LiPo battery (example)	1	11.1 V · 2200 mAh · 25C continuous · ≈190 g

Appendix B — RoboReachProblem Statement Summary

This is a condensed version for your reference. Details marked "≈" may differ by about 10% on the real arena. Read it fully; interview questions can come from any of it.

B.1 Bot and run rules

Topic	Rule
Size	Fits in $300 \times 200 \times 300$ mm (l × b × h) at start; may extend after the run begins.
Arm	Mounted on the bot; must grasp objects with centre at 450 mm height. Ground clearance ≥ 50 mm.
Gripper	Mechanical only. No adhesives, magnets, hooks, suction cups. No ready-made Lego; ready-made gear assemblies allowed.
Drive	Wheels in continuous ground contact.
Control	Manual only; one operator; no autonomous navigation or manipulation. Allowed: Wi-Fi, Bluetooth (Classic/BLE), 2.4 GHz FHSS RC. Phone joystick allowed; no internet or cellular.
Authentication	The bot must accept only its team's controller and reject or ignore all others.
Disconnect	Motors stop, arm stops, gripper stays at previous position, bot stays disabled until a new valid command.
Power	Onboard supply only; ≤ 24 V DC between any two points; battery securely mounted; kill-switch that shuts everything down; no AC.
Run	420 s (7 min), one run. Score $= A + T - P$. Time bonus $T = 300 -$ time(s) if time < 300 s, capped at 100. Skipping an obstacle earns no points for it but saves time.
Objects	Made of thermocol or equivalent. Arena $\approx 3 \times 3$ m.

B.2 Obstacle sequence and points (maximum 870 + 100 time bonus)

#	Obstacle	Key details	Points
1	The Pickup	Lift and hold a 50 mm cube clear of the ground	10
2	The Ascent	Ramp with $15°$ incline, holding the cube with the arm only	30
3	The Bridge	Cross a 400 mm-wide bridge, still holding the cube	10
4	The Descent Breakdown	Descend and immediately cross speed breakers (peak 40 mm), holding the cube	60
5	Touchdown	Put the cube in an 80×80 mm box	10
6	Stacker	Stack three cylinders ($\varnothing 60 \times 80$ mm) and a cone (r 30, h 100 mm) in a 100×100 mm area. Tower must stand 5 s.	90 (20 per cylinder, 30 cone)
7	Longshot	Throw a 50 mm cube from behind the throw line. Lines every 10 cm	10 per line, max 100
8	Robonardo DaVinci	Pick up a marker and trace the "TF" logo inside the printed outline	130
9	Downy	Take a 60 mm cube from a height and place it at 400 mm height in the marked area	30
10	Hanoi'd	Solve Tower of Hanoi with three numbered 50 mm cubes on three 100×100 mm squares	80
11	Buzzed	Carry a loop along a buzz wire (up to 400 mm high) without touching it	150
12	The Diagonals	Obstacles narrow the track to 250 mm; tests turning in a tight space	20
13	The Escape	From a 600×600 mm box, identify 4 objects and drop each through its shaped hole	10 / 20 / 30 / 40
14	The Slalom Ridge	Zig-zag overlapping triangular barriers, carrying an object	40
15	The End	Whole bot stops in a 350×250 mm zone	10

B.3 Penalties

Event	Penalty
Any wheel leaves the track (out of bounds)	10 per violation
Cube dropped between Pickup and Touchdown	15 per drop
Object not completely inside the designated area at task completion	10
Longshot: crossing the throw line before release	10
Robonardo DaVinci: any part of a letter outside the outline	30
Downy: cube dropped between pickup and placement	10 per drop
Hanoi'd: any violation of Tower of Hanoi rules	20 per violation
Buzzed: loop touches the wire	20 per violation
Escape: object dropped before the box / failing to drop it through the hole	15 per drop / 20 per failure

Disqualification: failing any bot specification, damaging the arena, or interfering with another team's robot or wireless link.